

FACUNDO JOEL ALLIA FERNANDEZ

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Data Scientist and Python Developer with 7+ years bridging advanced analytics, machine learning, and financial engineering. Specialized in building production-grade data solutions, from ETL pipelines and automated systems to pricing models and risk analytics. Expert in Python, SQL, and cloud platforms with proven track record transforming complex financial data into actionable insights across Big Four firms, investment banking, and derivatives markets. Combines deep quantitative finance expertise with modern ML/AI implementation and scalable data architecture.

WORK EXPERIENCE

Banco Santander Corporate & Investment Banking – Senior Data Scientist & Automation Engineer | 2025 – Present

- Design and implement scalable data solutions combining Python, SQL, and cloud technologies to optimize financial processes and reduce operational costs.
- Develop automated ETL pipelines and data integration workflows from multiple sources including enterprise systems, data lakes and financial platforms.
- Build end-to-end applications and analytical tools with interactive dashboards, enabling data-driven decision-making.
- AI Ambassador promoting AI technology adoption across organization, developing use cases and facilitating implementation of AI solutions within business units.

Technologies: Python, SQL, Databricks, Murex, Power BI, RPA, Azure, ETL/ELT pipelines, RAGs, MCPs, LLMs, model deployment

PricewaterhouseCoopers (PwC) - Quantitative Analyst | 2024 – 2025

- Specialized in complex derivatives pricing using stochastic models and Monte Carlo simulation. Developed credit derivatives valuation engine with Python implementing CVA/DVA frameworks.
- market risk assessments and built XVA calculation models for regulatory compliance, collaborating with audit teams on validation frameworks.
- Regular contributor to technical newsletter analyzing Decentralized Finance trends and their impact on traditional markets, leveraging blockchain analytics and quantitative research.
- Performed scenario analysis and stress testing for financial instruments using Python, R, and proprietary risk systems.

Technologies: Python, R, Databricks, ML Models, PySpark, Power BI, Azure, Murex, Bloomberg Terminal, Reuters

Accenture - Financial Data Transition Specialist | 2022 – 2023

- Developed KPIs and financial metrics for enterprise reporting, identifying process optimization opportunities boosting profitability.
- Implemented forecasting tools and management software improving financial visibility and decision-making.
- Worked with SAP HANA and cloud platforms for large-scale data integration and analytics.

Technologies: Python, AWS, SAP, SAP HANA, SQL, ETL, Azure, Microsoft Excel, SharePoint, Databricks, PySpark

Nixtla - Data Scientist | 2022

- Performed time series forecasting and modeling (ARIMA, ETS, CES) using Python and R for predictive analytics.
- Developed automated forecasting pipelines on Databricks with PySpark for scalable model deployment.
- Authored technical documentation and conducted model performance evaluations ensuring optimal accuracy.

Technologies: Python, R, SQL, PySpark, Databricks, AWS, Docker, APIs, CI/CD, MLOps, Git/GitHub

Matba Rofex - Institutional Derivatives Trader - Market Making | 2021 – 2023

- Actively traded futures and options on financial assets and commodities, managing market-funded account to meet volume standards.
- Demonstrated strong quantitative skills forecasting market trends, **consistently outperforming benchmark for two consecutive years**.
- Developed Python-based trading analytics and backtesting frameworks for strategy optimization.

Technologies: Python, SQL, APIs, Trading platforms (Matriz, Etrader), TradingView

Logos Servicios Financieros - Financial Advisor & Founder | 2020 – Present

- Provide financial advisory services and portfolio management with customized investment strategies.
- Established and managed proprietary client portfolio ensuring high-quality service and satisfaction.

Libertad S.A. - Commercial Data Analyst | 2021 – 2022

- Developed monthly/annual budgets and demand forecasts incorporating macroeconomic variables via interactive dashboards.
- Implemented real-time sales data ingestion and analysis solution at scale, adding significant operational value.

Technologies: Python, SQL, SAP, Power BI, Microsoft Excel, proprietary ERP/CRM

Peugeot Avant Autocity - Data Analyst | 2019

- Maintained and structured operational databases using SQL, Excel, SAP and proprietary ERP/CRM systems.
- Built dashboards and performance reports supporting business decisions.

Technologies: SQL, Microsoft Exel, SAP, Power BI, proprietary ERP and CRM.

STUDIES AND CERTIFICATIONS

- **Master's in Finance -** EAE Business School Madrid (2024) - *Minor in Smart Money & Start Ups*
- **Postgraduate Degree in Quantitative Finance** (Quant Program) - UCEMA (2022)
- **Data Science Coding Bootcamp** (+700h.) - Henry (2022)
- **Financial Markets Specialization** (Derivatives trading focus) - Bolsa de Comercio de Rosario (2021)
- **Bachelor of Business Administration** (Finance Specialization) - Universidad Nacional de Córdoba (2020)
  Teaching assignment in Stock Exchanges and Stock Market
- Financial Market Suitability and Stock Market Producer Registration Nº 1271 - CNV (2021)

Courses and certifications:

- Cloud Practitioner Essentials - AWS
- Strategic thinking - CAME
- Personal brand - Platzi
- Harvard Negotiation Method - IEP
- Digital Marketing Strategies - CAME
- Python for Finance, Investment & Data Analytics – Udemy
- Time Series Forecasting - Kaggle
- Digital Disruption in Finance – UNC
- Regulatory Consulting – PWC
- PLAFT- Balanz University
- Personal finance - UNAM
- Advanced Excel - IPAP

TECHNICAL SKILLS

- Python: Numpy, Pandas, SciPy, and visualization libraries.
- Machine Learning: TensorFlow and Scikit-Learn.
- PySpark
- C++
- Git - GitHub
- Blockchain, cryptocurrencies, and DEFI experience
- Expert in Microsoft Office
- Knowledge of probability and statistical methods
- Financial Valuation
- Bloomberg Terminal
- SQL
- Amazon Web Services
- Power BI
- Databricks
- APIs, CI/CD, MLOps
- Docker
- Financial modeling (TSF, risk, etc.)
- R language
- LLMs, MCPs, agents, RAGs and AI tools
- Reuters Terminal

LANGUAGES

- **Advanced English**
  - EF SET English Certificate 77/100 (C2 PROFICIENT) - EF Standard English Test
  - Business English Certificate Vantage (CEFR LEVEL B2) - Cambridge University
- **Basic Portuguese** (A2) - Business oriented
- **Native Spanish**

SOFT SKILLS

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- Critical analysis and problem-solving skills
- Effective communication and negotiation skills
- Ability to work effectively under pressure
- Self-learning and continuous skill development
- Collaborative and cooperative teamwork
- Meticulous attention to detail
- Adaptability to changing situations
- Results and goal-oriented mindset

PUBLICATIONS

- Allia Fernandez, F. J. “Construction of risk-neutral probability density functions through butterfly spreads”. BCR. Jun 30, 2022. Available at SSRN
- Allia Fernandez, F. J. “Stock classification using k-means clustering”. UCEMA. Dec 7, 2022. Available at SSRN